

STATIC GRADIENT ATTRIBUTION UNDERPERFORMS A DENSITY-MATCHED RANDOM MASK ON LOSS-BASED FORGETTING WITHIN LoRA’S B-MATRIX

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ABSTRACT

Language models in the low-billions parameter range are increasingly deployed on edge devices, where fixed compute budgets must continuously absorb new domain tasks via parameter-efficient adapters (e.g., LoRA) without catastrophic forgetting. Continual learning typically restricts updates to a parameter subset chosen by gradient attribution—using a calibration backward pass to score importance. We show that previous evaluations of this approach suffer from a fundamental confound: under standard zero-initialized LoRA ($B = 0$), the gradient with respect to A is identically zero ($\partial\mathcal{L}/\partial A = 0$), forcing gradient masks to select strictly from the B matrix, whereas standard random baselines draw from both A and B . The comparison these evaluations run is itself null (Backward Transfer 0.437 ± 0.061 versus 0.428 ± 0.019 , $t(4) = 0.41$, $p = 0.70$); matching the parameter pool is what surfaces the effect. Isolating the selection criterion by restricting both approaches to the same parameter pool (10% of LoRA B , 1.65M parameters), we find a negative result: a uniform random mask consistently achieves lower loss-based backward transfer than gradient attribution across seven seeds on Qwen2.5-3B (mean BWT 0.294 vs. 0.449, $\Delta = -0.155$ [Random – Grad] on the V100 stack, $p = 0.0011$; 7/7 seeds). The effect is concentrated on the medical task, which contributes 80% of the aggregate; code and math move in the same direction, and legal shows no reliable difference. This advantage is consistent with a stability–plasticity trade-off, in which random masking slightly reduces single-task fitting. On accuracy the two masks are statistically indistinguishable on the two discrete-option tasks (Medical, Legal) (+0.94 pp, 95% CI $[-2.91, +4.80]$). A 4-percentage-point non-inferiority margin specified in advance is not met (upper bound +4.80 pp). On those same two tasks, accuracy differences of up to about 4.8 percentage points in the gradient mask’s favour are not excluded by this measurement. On loss-based forgetting, which is the metric this paper measures, gradient attribution is measurably worse than density-matched random masking, and the attribution pass—adding per-task backward overhead and calibration storage—can be dropped in favour of a simpler, zero-cost alternative for resource-constrained LLM continual learning.

1 INTRODUCTION

Language models in the low-billions parameter range are increasingly fine-tuned with parameter-efficient adapters (Hu et al., 2021; Dettmers et al., 2023), where a fixed compute budget must absorb a stream of new tasks while reducing catastrophic forgetting. Continual learning for LoRA adapters updates only a chosen subset of parameters, and gradient attribution is the common rule for choosing it: a backward pass over held-out calibration data scores each parameter, and the highest-scoring ones are updated. Choosing the subset uniformly at random costs nothing instead, so whether the attribution pass earns its cost has a direct engineering answer.

The question is harder to settle than it looks, because standard zero-initialized LoRA introduces a confound. With $B = 0$ at initialization, $\partial\mathcal{L}/\partial A = 0$, so gradient attribution assigns zero im-

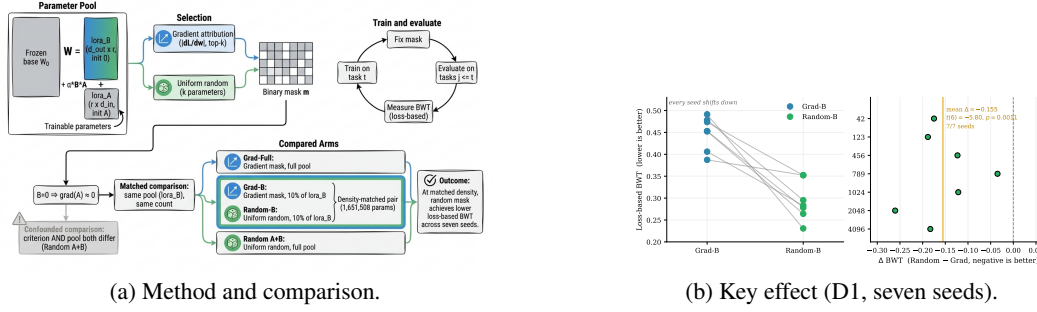


Figure 1: The density-matched comparison. (a) Both arms draw 10% of the rank-16 LoRA LoRA_B pool, so they train exactly the same 1,651,508 parameters and differ only in how those coordinates are chosen; because $B = 0$ at initialization, the gradient score is zero for every LoRA_A coordinate, and any baseline that draws from both pools changes the pool as well as the criterion. (b) Under that matched density the random mask achieves lower loss-based BWT than the gradient mask in all seven seeds (mean $\Delta = -0.155$, $p = 0.0011$).

portance to every LoRA_A parameter and a gradient mask selects only from LoRA_B , while the usual random baseline draws from both factors together. Any comparison that does not match the two pools therefore mixes the *selection criterion* with the *parameter pool*; the closest prior work, Attribution-Guided CL (Liu et al., 2026), uses gradient attribution as a continual-learning mask on LLMs without matching the pool.

We separate the two by restricting both arms to the same pool and parameter count. On Qwen2.5-3B with rank-16 LoRA over a five-task sequence we compare a uniformly random mask on LoRA_B (Random-B) against a gradient mask renormalised to the same 10% of LoRA_B (Grad-B), so both train exactly the same parameters and differ only in how those parameters are chosen. Across seven seeds the random mask achieves the lower loss-based backward transfer, and the gap holds in every seed (Table 2, Figure 1). The aggregate gap is concentrated on the medical task, with code and math moving in the same direction and legal showing no reliable difference, a pattern consistent with a stability–plasticity trade-off: the random mask fits each task less well on its own diagonal yet retains earlier tasks better.

Contributions.

1. **A gradient-attribution confound in LoRA continual learning.** Under zero-init LoRA the gradient with respect to LoRA_A is identically zero, so gradient masks select from LoRA_B only while the usual random baseline draws from both pools. Comparisons that do not match the pool compare a selection rule against a selection rule plus a pool change.
2. **Density-matched evidence that random selection is not worse.** At equal parameter count the random mask achieves the lower loss-based BWT in every seed, an advantage concentrated on the medical task, with code and math in the same direction and no reliable difference on legal. The pattern is a stability–plasticity trade-off, not a uniformly better mask.
3. **An engineering recommendation: drop the attribution pass.** At matched density a zero-cost random mask is not worse on loss-based forgetting. On the two discrete-option tasks (Medical, Legal), the two masks are statistically indistinguishable, and a 4-percentage-point non-inferiority margin specified in advance is not met: accuracy differences of up to about 4.8 percentage points in the gradient mask’s favour are not excluded by this measurement. The recommendation is therefore scoped to the loss evidence, and it removes the per-task backward pass over the calibration set (64 forward–backward steps, 12.8% on top of the training budget) together with the calibration-set dependency.

All reported forgetting metrics are loss-based BWT. All headline comparisons are within-run.

Table 1: Notation. Indices run over tasks $t = 1, \dots, T$ with $T = 5$, and over prior tasks $j \in \mathcal{P} = \{1, \dots, T - 1\}$.

Symbol	Meaning
f_θ, W_0	base model and its frozen pretrained weight matrix
B, A	low-rank LoRA factors, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$
r, α	LoRA rank ($r = 16$) and scaling ($\alpha = 32$)
W	adapted weight, $W = W_0 + \alpha BA$
t, T	task index and number of tasks ($T = 5$)
$\mathcal{D}_t$	training split of task t
$\mathcal{D}$	held-out calibration set (64 Alpaca samples)
m	binary mask over the trainable coordinates, $m \in \{0, 1\}^{ A + B }$
k	selection budget: number of coordinates with $m = 1$
g_i	gradient-attribution score of coordinate i (§3.3)
$\mathcal{L}_j^{(\tau)}$	evaluation loss on task j after training through task τ
$\mathcal{P}$	prior-task set $\{1, \dots, T - 1\}$
BWT	loss-based backward transfer, Eq. equation 1

2 BACKGROUND

Continual learning (CL) trains a model on a sequence of tasks without catastrophic forgetting (McCloskey & Cohen, 1989; French, 1999). A dominant strategy identifies important parameters and protects them during later tasks: EWC (Kirkpatrick et al., 2017) uses the diagonal Fisher information as a quadratic penalty, SI (Zenke et al., 2017) accumulates path-integral importance, and gradient-based selective-update methods score parameters from calibration gradients, an idea since extended to LLMs and parameter-efficient tuning (Qiao et al., 2024; Wang et al., 2025a). Its cost is the attribution pass’s transient backward peak, which a random within-lora-B mask removes entirely (Appendix B).

The alternative view is that the identity of the selected parameters matters less than how many are selected. The lottery ticket hypothesis (Frankle & Carbin, 2019) and supermasks (Zhou et al., 2019; Ramanujan et al., 2020) show random subnetworks in large networks can perform strongly, Aghajanyan et al. (2021) and Li et al. (2018) show that fine-tuning occupies a low-dimensional subspace, and DARE (Yu et al., 2024) shows that most delta parameters can be dropped. The same theme appears in pruning, where Liu et al. (2019) show pruned architectures can be retrained from scratch and Blalock et al. (2020) find that method choice matters less than the sparsity budget. LoRA already constrains updates to a low-rank subspace and masking restricts further within it, so the question is whether attribution beats a random restriction of the same size.

3 METHODOLOGY: MASKING MECHANISMS

3.1 PROBLEM FORMULATION

Continual learning trains a model on a sequence of tasks without losing performance on earlier ones (McCloskey & Cohen, 1989; French, 1999). We use the standard sequential fine-tuning protocol: a base model f_θ is adapted to task t on its training split $\mathcal{D}_t$, then evaluated on all tasks $j \leq t$. Tasks arrive in a fixed order indexed $t = 1, \dots, T$, with $T = 5$ throughout this paper. An adapter adds a low-rank update $\Delta W = BA$ to a frozen weight matrix W_0 , where $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $r \ll \min(d_{\text{out}}, d_{\text{in}})$, and $W = W_0 + \alpha BA$. Table 1 collects the notation used from here on.

Input and output. The method takes four inputs: the pretrained base model, the ordered stream of T tasks with their training splits, a small held-out calibration set drawn from a different distribution, and the selection budget k , the number of coordinates the mask may activate. It returns a single binary mask, fixed before the first task and shared by all of them, together with the adapter produced by training the unmasked coordinates under that mask. The mask is the object of study; nothing about it is task-specific or learned after the first forward pass over the calibration set.

Parameter masking applies a binary mask $m \in \{0, 1\}^{|A|+|B|}$ to the trainable parameters, confining updates at task t to coordinates where $m = 1$. Two selection rules are compared: *gradient attribution*, which scores each coordinate by its calibration-set gradient magnitude, and *uniform random selection*. Both hold the budget k fixed, so the two arms train the same number of parameters and differ only in which ones they are.

We measure forgetting by loss-based backward transfer. The backward transfer is

$$\text{BWT} = \frac{1}{|\mathcal{P}|} \sum_{j \in \mathcal{P}} \left(\mathcal{L}_j^{(T)} - \mathcal{L}_j^{(j)} \right), \quad (1)$$

the mean increase in evaluation loss on the prior tasks $\mathcal{P} = \{1, \dots, T-1\}$ after the full sequence; lower values mean less forgetting. Section 4.2 defines the metric and the accuracy criterion in full.

3.2 THE B=0 GRADIENT ATTRIBUTION CONFOUND

As set out in §1, standard zero-init LoRA has $B = 0$ at initialization, so gradient attribution assigns zero importance to every `lora_A` parameter and the mask selects exclusively from `lora_B`. This subsection gives the control measuring the consequence.

This asymmetry is a control experiment for the within-`lora_B` comparison. Restricting gradient attribution to `lora_B` is a no-op by construction and leaves the measured BWT identical, so the gradient mask is already confined to `lora_B` without any explicit restriction; the same restriction applied to the random arm instead removes its `lora_A` dilution and improves it (Table 13, Appendix O). That the restriction helps one arm and not the other is the direct evidence that the within-B comparison varies the selection criterion rather than the pool. Attribution-guided mask selection (Liu et al., 2026) therefore inherits a pool mismatch whenever its baseline draws from both pools, and we run that comparison matched in §5.2.

We scope this B=0 effect to *standard* zero-initialized LoRA. PiSSA (Meng et al., 2024) initializes both A and B from principal singular values, so $B \neq 0$ and the gradient mask does not collapse to `lora_B`-only; LoRA+ (Hayou et al., 2024) addresses the related asymmetric A/B dynamics. Whether attribution helps under non-zero initialization is open.

3.3 DENSITY-MATCHED MASK CONSTRUCTIONS

A selection rule scores every trainable coordinate and keeps the top k of them. Gradient attribution scores coordinate i by the mean absolute calibration-set gradient,

$$g_i = \mathbb{E}_{x \sim \mathcal{D}} \left[\left| \frac{\partial \mathcal{L}(x)}{\partial w_i} \right| \right], \quad (2)$$

and the mask keeps the k largest scores,

$$m_i = \mathbb{I}[g_i \geq \tau_k], \quad \tau_k = \max \{ \tau : |\{i : g_i \geq \tau\}| \geq k \}, \quad (3)$$

so that exactly k coordinates are active and ties go to the higher score. Uniform random selection replaces g_i by a draw from $\text{Uniform}(0, 1)$ and applies the same top- k rule, which is what makes the two arms comparable: same budget, same pool, only the ranking rule differs.

The pools themselves are not interchangeable, and matching them is the point of the control. Let $\mathcal{B}$ denote the `lora_B` sub-pool, $|\mathcal{B}| = 29,933,568$. The unrestricted gradient mask applies equation 3 over all trainable coordinates and, because $B = 0$ makes equation 2 vanish on every `lora_A` coordinate (§3.2), it in fact selects a subset of $\mathcal{B}$ alone. The density-matched gradient arm instead renormalises the budget to the same fraction of $\mathcal{B}$ the random arm uses,

$$k_{\text{Grad-B}} = \lfloor 0.1 |\mathcal{B}| \rfloor = 1,651,508 = k_{\text{Random-B}}, \quad (4)$$

so density and pool are held fixed and the selection criterion is the only free variable.

We compare seven strategies, in the short forms used throughout: *Grad-Full* for the unfiltered gradient mask, *Grad-B* for the density-matched gradient mask restricted to `lora_B`, and *Random-B* for the uniform random mask on the same pool. (1) **Static OOD-calibrated gradient mask** (Grad-Full) scores by equation 2 over 64 *out-of-domain* Alpaca samples and selects the top-10% once for

all five tasks; by the $B = 0$ effect (§3.2) it draws only from `lora_B`. (2) **Gradient B-only control** (`full_lora_B_only`) is (1) with the attribution pool restricted to `lora_B`, a null check *identical* to (1) at all seven seeds. (3) **Density-matched gradient arm** (Grad-B) is the headline arm: the budget is renormalised by equation 4 so that 10% of the `lora_B` sub-pool is selected (1,651,508 parameters); together with (5) it forms D1 (Table 2). (4) **Random mask (A+B)** draws 10% uniformly at random from all LoRA parameters, and (5) **Random B-only** (Random-B) draws 10% uniformly at random from `lora_B` only, the density-matched partner of (3). These three `lora_B`-restricted constructions must not be conflated: (2) is a no-op equalling the confounded mask, (3) is the density-matched headline arm, and (1) is the confounded arm at full-pool top-10% (2,993,357), so only (3) is density-matched. (6) **No mask** is standard LoRA with all parameters updated, and (7) **uniform B** spreads the 10% budget uniformly across LoRA modules (soft 0.7 shrinkage) while drawing from `lora_B`; the latter is seed-dependent and does not establish the mechanism (§6.2).

Algorithm 1 (Appendix A) states the end-to-end procedure. The mask is built once from the calibration set before any task is trained, then frozen; the task loop trains only the coordinates where $m = 1$, so no step of the procedure is task-specific.

4 EXPERIMENTAL SETUP

4.1 MODELS, BENCHMARKS & TRAINING DETAILS

Qwen2.5-3B (Team, 2024) in 4-bit NF4 (bitsandbytes), LoRA rank 16 ($\alpha=32$), AdamW ($\text{lr } 2 \times 10^{-4}$, warmup 0.03), 1 epoch, 500 train samples per task, batch size 1, max length 512; calibration uses 64 Alpaca samples. All methods operate on 29.9M trainable LoRA parameters (0.96% of 3.1B); the full-pool top-10% selects 2,993,357, falling entirely within `lora_B` ($B = 0$) and activating $\sim 20\%$ of B, against 10% for Random-B. The comparison is matched on pool but not density; §5.2 equalizes it.

Five domains are used in a fixed order: Code (sahil2801/CodeAlpaca-20k), Medical (GBaker/MedQA-USMLE-4-options), Legal (lex_glue, casehold subset), Math (openai/gsm8k), Creative (ChaoticNeutrals/Creative_Writing-ShareGPT, 6.6K instruction-response pairs). Calibration: tatsulab/alpaca (64 samples); 100 eval samples per task.

4.2 EVALUATION METRICS

Loss-based BWT is the average increase in evaluation loss on prior tasks after learning new ones, lower meaning less forgetting, as defined in §3. We use it in preference to accuracy-based BWT because loss is continuous and more sensitive to the small differences between masking strategies. All reported forgetting metrics are loss-based, reported as mean $\pm$ std over seven seeds (42, 123, 456, 789, 1024, 2048, 4096); part of the runs used a second GPU. The accuracy side of the same comparison, together with the option log-likelihood scorer and its scope, is reported separately in §5.4.

5 EXPERIMENTS AND MAIN RESULTS

5.1 CONFOUNDED AND DENSITY-MATCHED COMPARISONS

A first comparison follows the usual design, in which each method draws from its own pool. Under that design gradient attribution shows no advantage: its loss-based BWT is 0.437 ± 0.061 against 0.428 ± 0.019 for a shared random mask over five seeds, a seed-dependent difference that is not significant (Table 22, Appendix W). Both methods still reduce forgetting relative to an unmasked adapter, but the comparison is uninformative about the selection rule, for the reason set out in §3.2: gradient attribution selects only `lora_B` while the random baseline draws from both pools, so the design varies pool and criterion together. The result is reported for completeness and is distinct from the seven-seed comparison that follows, the two coming from different sessions whose absolute levels are not comparable (Appendix F).

Holding the pool fixed changes the picture. We introduce a Random-B control drawn from `lora_B` alone, matching the gradient mask’s pool, and compare it against a gradient mask restricted to the

Table 2: **Headline result (D1, density-matched).** Both arms select the same 1,651,508 parameters (10% of `lora_B`); only the criterion differs. Per-seed loss-based BWT and $\Delta = \text{Random-B} - \text{Grad-B}$ (negative = random-B forgets less), seven sessions, one seed each. The last column is the density-*unmatched* contrast from the same sessions, that is, Random-B against the unrestricted gradient mask rather than against Grad-B.

Seed	Random-B BWT ↓	Gradient-B BWT ↓	Δ (matched)	Δ (unmatched)
42	0.278	0.453	−0.175	−0.220
123	0.264	0.453	−0.188	−0.119
456	0.283	0.406	−0.123	−0.131
789	0.352	0.387	−0.035	−0.074
1024	0.352	0.474	−0.121	−0.109
2048	0.231	0.491	−0.261	−0.187
4096	0.295	0.478	−0.183	−0.166
Mean	0.294	0.449	−0.155	−0.144

same pool. Each seed is an independent session, so absolute BWT values drift between runs (Appendix F) and only within-session paired observations are used. This comparison is matched on pool but not on density, since the gradient mask activates nearly twice as many `lora_B` coordinates as Random-B. Even so, the random mask achieves the lower loss-based BWT in all seven seeds, with a mean difference of -0.072 that survives dropping the largest-effect seed (paired $t(5)$, $p = 0.015$; Figure 2, Appendix E). That the direction is unchanged when the pool is fixed is the first indication that the criterion, rather than the pool, is what the earlier comparison was measuring.

5.2 DENSITY-MATCHED CONTROL

Pool and criterion are separated cleanly only when density is matched as well. The comparison above leaves the gradient arm activating about 20% of the pool against 10% for Random-B, so the two arms still differ in how many coordinates they train. D1 removes that difference, with both arms drawing exactly 10% of `lora_B` and only the ranking rule differing. At matched density the random mask again achieves the lower loss-based BWT in every seed, by a larger margin than in the density-unmatched comparison (Table 2); it reproduces at the same magnitude on a second GPU, and the direction holds under a reversed task order.

5.3 TASK-LEVEL DECOMPOSITION

The aggregate gain is not spread evenly across the five tasks, and where it concentrates bears on how the effect should be read. Medical carries about four fifths of it, while code and math contribute the remainder in the same direction; on legal the arms are indistinguishable. The concentration is strong but not total, since removing the medical task leaves the matched contrast at a little over a quarter of its headline size and still negative in every seed. The three-task test is directional only, and the seven-seed test carries the significance claim (Appendix U).

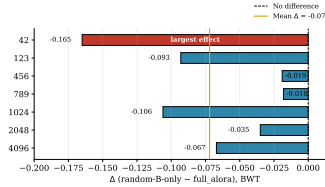


Figure 2: Per-seed within-`lora_B` differences ($\Delta = \text{Random-B} - \text{Grad-Full}$, loss-based BWT, 7 seeds). All seven are negative (Random-B forgets less), mean $\Delta = -0.072$ (orange line); seed 42 is the largest effect (-0.165). Leave-one-out excluding seed 42 remains significant (paired $t(5)$, $p = 0.015$).

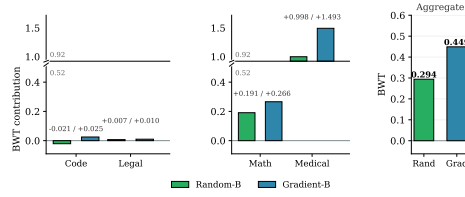


Figure 3: D1 per-task decomposition, seven-seed means. Each task shows its Random-B (green) and Grad-B (blue) loss-based BWT contribution; the label above each task gives those two values in the same order. The per-task contributions span roughly two orders of magnitude, so the three small-magnitude tasks (Code, Legal, Math) share the lower segment of a broken y-axis while Medical is carried in the upper segment; the pair of slashes on each spine marks the omitted band, and the shared zero line runs unbroken across both segments. The right panel repeats the aggregate, a different quantity, on its own scale, drawn with its zero line on the same baseline. Random-B is lower on every task, with the effect concentrated on Medical; the per-seed table is in Appendix U.

Two rival explanations for the effect can be excluded from the same data. The gradient arm’s loss on each task’s own diagonal is lower than the random arm’s in every seed and task, yet it forgets more, so the two arms differ in retention rather than in learning capacity (Appendix V). A second reading, that the gradient arm would do better with its full budget, rests on a single seed: at seed 42 the density-matched gradient arm beats the unrestricted one, but at the seven-seed mean the ordering reverses.

5.4 ACCURACY-BASED VALIDATION

Loss-based BWT is a proxy for forgetting rather than the quantity of ultimate interest, so we evaluate the same masking comparison on task accuracy and ask whether the accuracy ordering agrees with the loss ordering. The evaluation reuses the primary protocol and the corrected task evaluators over the same seven seeds and 100 evaluation items per task, retraining from the base model on an A800 rather than re-scoring a saved checkpoint. Accuracy is scored only for the two tasks whose answers are discrete options (Medical, MedQA four options; Legal, CaseHOLD), so every quantity below concerns those two.

The two masks are indistinguishable on accuracy: across seven seeds their difference averages $+0.94$ percentage points with a 95% interval of $[-2.91, +4.80]$ that spans zero. A non-inferiority margin of 4 percentage points was specified in advance for this interval, and the measured upper bound exceeds it, so that criterion is not met. On the loss side the same masks are separated, with the gradient mask forgetting more in all seven seeds by a mean of $+0.152$; measured a second time on the A800 under a different implementation with the loss metric defined identically, the two estimates agree in magnitude to 0.003 once the sign convention is aligned (the V100 figure is reported as [Random – Grad] and the A800 figure as [Grad – Random]; both describe the same contrast). Loss therefore resolves a difference the accuracy measurement does not, and the accuracy interval bounds that difference rather than contradicting it (Figure 4; per-seed values and the interval construction in Appendix P).

One task is excluded from the accuracy read-out. Loss on math decreases with training while accuracy on the same task collapses to zero for every arm, a consequence of truncation rather than of the masking choice and common to all arms, so we do not treat the math column as a forgetting result; the audit is in Appendix R.

6 ANALYSIS AND EXTENSIONS

6.1 SPARSITY, SCALE, AND CROSS-ARCHITECTURE GENERALIZATION

The headline comparison fixes the sparsity budget at 10%. Sweeping it from 5% to 30% at a single seed, no tested level shows gradient attribution outperforming a shared random mask (Figure 6, Appendix H). The sweep is preliminary and inherits the pool confound rather than isolating the

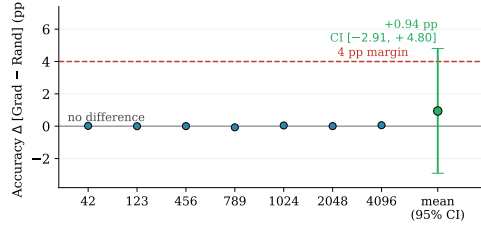


Figure 4: Accuracy-based validation. Blue points are the seven per-seed accuracy differences (gradient minus random, percentage points) on the two discrete-option tasks; green is the seven-seed mean with its 95% interval $[-2.91, +4.80]$ pp. The interval spans zero, so the masks are indistinguishable on accuracy, and it reaches past the 4 pp non-inferiority margin, so that criterion is not met. Per-seed values in Appendix P.

budget, so we report it as a boundary check on the direction of the effect rather than as evidence for the sparsity-constraint view.

Boundaries in scale and task order were probed next, followed by a second architecture. No single-seed probe reverses the conclusion, but a reversed task order shifts BWT by more than any method difference, which is why every conclusion in this paper is scoped to the tested sequence (Table 11, Appendices H–L). On Phi-2, an architecturally distinct family, the within-`lora_B` direction replicates under the density-unmatched protocol across four seeds, with an effect size comparable to Qwen2.5-3B. Two features of that replication matter for how it should be read: the two Phi-2 designs use different LoRA target sets and are reported separately rather than pooled, and Phi-2 forgets roughly $2.5\times$ less overall, so the magnitude is architecture-dependent even though the direction is not (Appendix Q).

6.2 MODULE CONCENTRATION AND CALIBRATION ROBUSTNESS

The gradient mask does not select uniformly across module types: within `lora_B` it concentrates on output projections and to a lesser degree on `up_proj`, together accounting for a little over half of all selections, while down projections receive well below the uniform rate. The pattern is consistent across calibration sources and peaks at middle layers (Appendix J). That concentration invites the reading that it causes the gradient mask’s higher BWT, and the `uniform_B` controller tests it by spreading the same budget across modules. The controller contradicts the reading: three of four seeds land on the gradient side rather than the random side. Because those offsets are taken against different session baselines (Appendix F), we treat the ablation as failing to establish the mechanism rather than as evidence against the concentration itself, and neither the within-`lora_B` finding nor its density-matched control depends on it (Figure 7). The result is also insensitive to the calibration domain: pooling in-domain rather than out-of-domain calibration changes 42% of the selected parameters, yet the loss-based BWT moves by less than a tenth of the between-seed spread (Appendix N).

7 RELATED WORK

Gradient- or saliency-scored importance is the shared assumption behind two otherwise separate lines. In regularization-based continual learning, EWC (Kirkpatrick et al., 2017), SI (Zenke et al., 2017) and MAS (Aljundi et al., 2018) score parameters by Fisher information or output sensitivity and penalize movement in the coordinates that score highest, and a separate line constrains updates to a gradient subspace (GPM (Saha et al., 2021), OGD (Farajtabar et al., 2020), GORP (Wang et al., 2025a), Model Shapley (Xu et al., 2025)). All of them assume that *which* parameters are protected matters, and so do we; the question we put to that assumption is whether the selection criterion within the trainable pool has any measurable consequence once the pool and the budget are fixed (§5.1, §5.2). This complements RepT (Li et al., 2026), which argues that parameter-space gradients are a noisy coordinate system for attribution.

Our setting fixes the chosen fraction and varies only the ranking rule over a shared pool, which isolates the criterion from the budget. Under one protocol we train and evaluate four rules, Grad-Full, Grad-B, Random-B and Random A+B, and compare the remaining rules by their published design only, with none of their reported numbers presented here as a measurement of ours; Table 17 in Appendix T lists them side by side. That separation also distinguishes rules deciding *which* parameters to protect from rules deciding *how many*: the pruning-at-initialisation scorers (SNIP (Lee et al., 2019), GraSP (Wang et al., 2020)), the one-shot pruning scorers (SparseGPT (Frantar & Alistarh, 2023), Wanda (Sun et al., 2024), movement pruning (Sanh et al., 2020)) and LoRA-FA (Zhang et al., 2023a) all fix their subset once and never revisit it, and pruning-at-initialisation scores are largely interchangeable with a per-layer choice (Frankle et al., 2021). Mask-learning variants instead adapt a fixed backbone per task (Mallya et al., 2018) and serve as CL task-specific subnetworks (Kang et al., 2022); Wanda importance is nearly orthogonal to gradient importance ($\rho = 0.086$).

Several methods show that parameter selection can improve continual learning: HAT (Serrà et al., 2018) learns per-task masks, Yang et al. (2025) reduce inter-task collision via sparse LoRA subspaces, Wang et al. (2025b) update only 0.1% of parameters per task, Lin et al. (2025) keep only uniquely relevant memory slots, Deng et al. (2024) revisit delta-parameter pruning, and Xu & Zhang (2024) show that random masking already discovers winning tickets for PEFT. Our finding does not contradict these; under our conditions the identity of the selected 10% matters less than the fact that 90% are frozen. Within parameter-efficient tuning, LoRA (Hu et al., 2021), QLoRA (Dettmers et al., 2023) and PackNet (Mallya & Lazebnik, 2018) constrain updates to parameter subsets, Li & Hoiem (2018) (LwF) uses distillation, and Biderman et al. (2024) find that LoRA forgets less than full fine-tuning. LoRA-FA (Zhang et al., 2023a) freezes `lora_A` and trains only `lora_B`, whereas we freeze a random or gradient-selected subset of an otherwise standard LoRA; other variants refine the parameterization (rsLoRA (Kalajdzievski, 2023), DoRA (Liu et al., 2024), AdaLoRA (Zhang et al., 2023b), LoRA+ (Hayou et al., 2024)) or enforce per-task subspace orthogonality (O-LoRA (Wang et al., 2023), InfLoRA (Liang & Li, 2024), Janus-LoRA (Chen et al., 2026), MoRAM (Lu et al., 2026), ASO-LoRA (Wang et al., 2026), SLoRA (Yang et al., 2026)). PiSSA (Meng et al., 2024) sets $B \neq 0$ at initialization, so the $B = 0$ mechanism our confound rests on is specific to standard zero-init LoRA, and most adjacent, Attribution-Guided CL (Liu et al., 2026) uses gradient attribution as a CL parameter-selection mask for LLMs, the class our within-`lora_B` finding evaluates. GaLore (Zhao et al., 2024) and LoGra (Choe et al., 2024) reduce gradient memory and scale data valuation through gradient projection, and Dieudonné et al. (2026), Adewuyi et al. (2026) and Azghan et al. (2026) report low-rank adapter redundancy, random sparse subnetworks for RLVR, and gradient-free latent routing respectively.

8 CONCLUSION

At equal parameter count within the `lora_B` pool, the random mask achieves the lower loss-based BWT in every seed, an advantage that reproduces on a second GPU and is not an artifact of density; the density-unmatched within-`lora_B` comparison points the same way, and the zero-initialization confound explains why earlier comparisons that did not match the pool failed to see it. The practical consequence follows from the cost structure this removes: the attribution pass answers a question a deployment setting re-asks whenever the task stream changes, namely which parameters should absorb the new task, and our measurement says the pass is not needed to answer it. Dropping it removes a per-task backward pass over the calibration set, 12.8% of the training budget in our configuration, and releases the calibration set with its activations, since nothing is scored. All comparisons are loss-based and scoped to the tested task order; a per-task adaptive mask or a non-NF4 quantization would be the most informative next test.

Limitations. The within-`lora_B` finding is bounded by several caveats, among them static masks only, absolute BWT that is not reproducible across sessions, a non-robust B-pool advantage, loss-based evaluation, a single task order, and one LoRA rank, each stated in Appendix S. Every claim rests on within-run comparisons, where the paired-comparison error is smaller than the positional and cross-session offsets that shift absolute levels (Appendix F); the output-projection suppression control is seed-dependent, so the concentration is an observation rather than a mechanism.

REPRODUCIBILITY STATEMENT

Experiments use Python 3.12, PyTorch 2.8.0+cu128 (4-bit NF4 with bf16 compute), Transformers 5.16.1, PEFT 0.20.0, bitsandbytes 0.50.2, datasets 5.0.1, and CUDA 12.8. The primary seven-seed within-lora_B comparison, scale runs, and ablations use a single NVIDIA A800 80GB PCIe GPU. The D1 density-matched control and the M2 uniform-B control ran in a separate session on a Tesla V100-PCI-E-32GB under an earlier environment (Python 3.12.3, PyTorch 2.3.0+cu121, Transformers 4.46.3, PEFT 0.13.2, bitsandbytes 0.43.3); as both are within-run paired comparisons, the hardware difference does not affect their paired results. Primary results use Qwen2.5-3B (seven seeds); cross-architecture within-B results use Phi-2 2.78B (four seeds); the confounded Phi-2 comparison uses five seeds under a matched 100-eval-sample protocol (an additional seed-42 observation used 20 eval samples and is reported separately); scale observations use Qwen2.5-0.5B/1.5B/7B (single seed). LoRA is applied to all linear projection layers.

Model training uses AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01) at learning rate 2×10^{-4} with a linear warmup of 0.03 and cosine annealing over the per-task step budget; LoRA dropout 0.05, rank 16, $\alpha=32$, batch size 1, max length 512, one epoch per task, and top- k 0.1 masking. Five sequential tasks are used: Code (code_alpaca_20k), Medical (medqa, 4-option MC), Legal (lexglue/case_hold), Math (gsm8k), and Creative Writing (Creative_Writing_ShareGPT). Each task uses 500 training and 100 evaluation samples; gradient masks use 64 out-of-domain Alpaca calibration samples. Seeds are 42, 123, 456, 789, 1024, 2048, and 4096; approximately 60 GPU-hours were used in total.

Boundary analyses (rank sweep, task order reversal, scale observations) are single-seed; a method-order check over three seeds is reported in Appendix M. An isolated-process protocol equivalence check confirms that running each method in a separate process reproduces the multi-method BWT to within 9.6×10^{-5} ($10\times$ margin below the pre-specified 10^{-3} threshold). All reported forgetting metrics are loss-based BWT. Code and configuration files are released as anonymized supplementary material (code_anonymous.zip): training and attribution scripts, five-task data preparation, mask generation, the analysis scripts computing BWT and the paired statistics, and pinned dependency manifests for both environments.

ETHICS STATEMENT

This work studies parameter masking for continual learning and does not involve human subjects or sensitive data. It does not introduce new capabilities or datasets that raise ethical concerns. On data sources and licensing: the Code task uses CodeAlpaca-20k (CC BY 4.0), the Math task uses GSM8K (MIT), and the Medical and Legal tasks use MedQA and CaseHOLD, which are distributed on Hugging Face with no affirmative license declaration, so we use them for research evaluation only. The Creative task uses a synthetic creative-writing set composed of model-generated outputs rather than user conversations. The Alpaca data used to calibrate the gradient masks is released under CC BY-NC 4.0, a non-commercial license; we therefore make no commercial-use claim for models trained with it. The finding that random selection outperforms gradient attribution within LoRA's B-matrix may reduce computational cost in continual-learning deployments by removing the need for gradient-based parameter attribution. We do not foresee negative societal consequences specific to this contribution.

AI USE DISCLOSURE

AI coding assistants were used for experiment-script development and debugging, and AI assistance aided literature search and manuscript drafting and revision. The research design, experimental methodology, statistical analysis, and scientific conclusions are the authors' own work. All reported results were produced by the described experimental pipeline without AI-generated synthetic data; all analyses and claims were verified by the authors.

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Algorithm 1 Masked continual fine-tuning with a frozen selection mask

Require: base model f_θ with frozen W_0 ; task stream $\{(\mathcal{D}_t)\}_{t=1}^T$; calibration set $\mathcal{D}$; budget k ; rule $\in \{\text{gradient, uniform}\}$

Ensure: frozen mask m ; adapted adapter $W = W_0 + \alpha BA$

- 1: $\mathcal{S} \leftarrow$ the (A, B) LoRA coordinates $\triangleright$ trainable coordinates
- 2: **if** rule = gradient **then**
- 3: **for** each coordinate $i \in \mathcal{S}$ **do**
- 4: $g_i \leftarrow \mathbb{E}_{x \sim \mathcal{D}} [\|\partial \mathcal{L}(x) / \partial w_i\|]$ $\triangleright$ Eq. equation 2
- 5: **end for**
- 6: **else**
- 7: $g_i \leftarrow \text{Uniform}(0, 1)$ for each $i \in \mathcal{S}$
- 8: **end if**
- 9: $m \leftarrow \mathbb{K}[g_i \geq \tau_k]$ $\triangleright$ top- k rule, Eq. equation 3
- 10: **freeze** m $\triangleright$ shared by all tasks; never updated
- 11: $W \leftarrow W_0$ $\triangleright$ standard zero-init LoRA, $B = 0$
- 12: **for** $t = 1$ to T **do**
- 13: train the coordinates with $m_i = 1$ on $\mathcal{D}_t$, updating only those
- 14: evaluate $\mathcal{L}_j^{(t)}$ for all $j \leq t$ $\triangleright$ every task seen so far
- 15: **end for**
- 16: **return** m , and BWT from Eq. equation 1 over $\mathcal{P} = \{1, \dots, T - 1\}$

A THE END-TO-END PROCEDURE

The notation is collected in Table 1 in the Methodology. Figure 1(a) shows the same procedure as a diagram; Algorithm 1 states it step by step.

B ATTRIBUTION COST

Table 3: Attribution cost on Qwen2.5-3B (BF16). Both columns are from a single *full-parameter* measurement run (`--include-frozen, --full-precision`); the exact correspondence and its limits are described below. Peak GPU is `torch.cuda.max_memory_allocated` converted to decimal GB (profiling fields `*_peak_alloc_mb = 13168 / 7102` MiB, i.e. 13.8 / 7.4 GB); Time is the per-task attribution wall-clock recorded by the run (584.9 s gradient / 12.1 s Wanda).

Method	Peak GPU (GB)	Time (s)	Backward?
Gradient mask	13.8 [†]	584.9	Yes
Wanda proxy	7.4	12.1	No
Shared random / RRM	0 [‡]	0	No
No mask	0 [‡]	0	No

[†]*Full-parameter* attribution configuration (`--include-frozen`, scoring all 3,085,938,688 base parameters); no method in this paper uses this configuration. The attribution actually used in the continual-learning experiments has a transient backward peak of ~ 0.73 GB (4-bit NF4, seq 512, 64 calibration samples, gradient checkpointing enabled; `overhead_mb = 699.43` MiB over the base model), and contributes 0 for random-B-only masking by construction (Table 22).

[‡]Zero by construction (no attribution pass is run), not a measured value.

Both measured columns of Table 3 come from a single full-parameter attribution profiling run (`--include-frozen, --full-precision`), so they are internally consistent; that run scores all 3,085,938,688 base-model parameters, a configuration no reported method uses. We describe the memory figure first, then the time figure.

Gradient attribution requires 13.8 GB peak GPU memory for Qwen2.5-3B (`max_memory_allocated`, 13168 MiB $\rightarrow$ 13.8 GB in decimal units). This figure is from the full-parameter attribution configuration, which scores all 3,085,938,688 parameters of the base model. The peak comprises four components (the resident base-model weights, the forward and backward activation graph over the full model, the full-parameter gradient accumulator, and the CUDA context), but these were not separately instrumented in the surviving run record, so

we report the total rather than a measured decomposition. The backward pass does traverse the base model: attribution computes $\partial\mathcal{L}/\partial w$ over calibration samples, so both the forward and backward graphs span the full parameter space rather than the 29.9M-parameter LoRA surface. This configuration peaks at 13.8 GB, of which $\sim 90\%$ is resident base-model weights (6.23 GB in BF16) plus the full-model gradient buffer (6.23 GB); these are costs the CL configuration does not pay. It is therefore an upper bound on a configuration no method uses, not a measure of attribution’s own overhead; the attribution actually used in the CL experiments peaks at ~ 0.73 GB of transient overhead above the base model, and it is that overhead the within-lora_B random baseline removes (Table 22).

The 584.9 s per-task figure is the wall-clock of that same full-parameter run (`full_alora_time_s` = 584.9 s). As a checkpointing-off contrast, an earlier LoRA-only attribution profiling run (4-bit, seq 1024, 256 calibration samples, gradient checkpointing disabled) reports 3,512 MiB of overhead, a different regime (checkpointing off, longer sequences) that should not be combined with the 0.73 GB figure above.

C PRUNING SALIENCY IS ORTHOGONAL TO GRADIENT IMPORTANCE

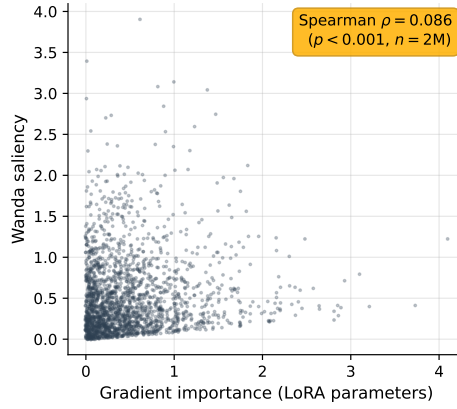


Figure 5: Spearman $\rho = 0.086$ between Wanda and gradient importance (LoRA parameters) on Qwen2.5-3B. These metrics rank parameters nearly independently.

Spearman correlation between Wanda ($|W| \times \text{RMS}(X)$) and gradient importance ($|\partial\mathcal{L}/\partial w|$) across 2M sampled parameters: (Figure 5; $\rho = 0.086$, 95% CI $\approx [0.084, 0.088]$, $n=2M$; correlation-only, measured outside a continual-learning context).

D CONFOUNDED NULL-RESULT STATISTICS

Table 4: Per-seed BWT breakdown (Qwen2.5-3B, rank-16 LoRA, 5 tasks).

Seed	Gradient BWT ↓	Random BWT ↓	No Mask BWT ↓
42	0.491	0.435	0.507
123	0.463	0.417	0.409
456	0.388	0.414	0.539
789	0.357	0.417	0.484
1024	0.486	0.458	0.479
Mean	0.437	0.428	0.484
Std	0.061	0.019	0.047

A paired t -test on per-seed BWT (Table 4; random vs. gradient: seeds 42/123/456/789/1024 yield differences $+0.056, +0.046, -0.026, -0.059, +0.028$) gives $t(4) = 0.41$, $p = 0.70$ (two-tailed), Cohen’s $d_z = 0.18$ (small effect). With $n=5$ and observed $\text{SD}_{\text{diff}} = 0.049$, the minimum detectable

Table 5: Within-`lora_B` comparison (Qwen2.5-3B, rank-16 LoRA, 5 tasks, 7 seeds). All comparisons are within-run (same method ordering and RNG state). $\Delta = \text{random_B_only} - \text{Grad-Full}$; negative Δ indicates random-B achieves lower (better) BWT.

Seed	random_B_only BWT $\downarrow$	full_alora BWT $\downarrow$	Δ
42	0.326	0.491	-0.165
123	0.320	0.413	-0.093
456	0.337	0.356	-0.019
789	0.346	0.364	-0.018
1024	0.336	0.442	-0.106
2048	0.367	0.402	-0.035
4096	0.331	0.398	-0.067
Mean	0.338 $\pm$ 0.015	0.409 $\pm$ 0.046	-0.072

BWT difference at $\alpha = 0.05$ is approximately 0.06 at 80% power, larger than the observed mean difference of 0.009. The direction is seed-dependent, consistent with no systematic advantage for either method. Normalizing BWT per task by zero-shot loss ($\Delta L/L_0$) does not change the method ranking. Tables 22 and 5 report different `full_alora` BWT for the same seed because they come from independent runs; the absolute BWT level varies across sessions (§F).

E WITHIN-`LORA_B` COMPARISON (DENSITY-UNMATCHED)

The within-`lora_B` comparison matches the parameter pool (both arms draw from `lora_B`) but *not* the density: the gradient mask activates approximately $2\times$ the B parameters of random-B-only. It is therefore an intermediate, density-unmatched step; the density-matched control (D1) is the headline comparison in the main text. All comparisons are within-run (same method ordering and RNG state); $\Delta = \text{random_B_only} - \text{Grad-Full}$, negative Δ favors random-B. Absolute BWT values carry session-level variation and should not be compared across seeds (§F).

F VARIATION ACROSS RUNS: A THREE-LEVEL DECOMPOSITION

Three quantities in this paper are sometimes read as competing estimates of one noise level. They are not: each measures a different thing at a different granularity, and the table below separates them.

Table 6: Three distinct quantities that are sometimes all called run-to-run *noise*. Only (a) is random error against a paired comparison; (b) and (c) are systematic offsets. They are not three estimates of one noise magnitude.

Level	What varies	Magnitude	What it measures
(a) within-session contrast noise	execution path only	9.6×10^{-5}	paired-comparison error; the only term the significance tests use
(b) position (ordering) offset	position of the two arms	0.064 (seed 42, <code>gradient_B</code>)	systematic; a position effect, not random error
(c) cross-session absolute offset	session and hardware	0.120 (<code>full_alora</code> , seed 42)	systematic; shifts the absolute BWT level between sessions

(a) Same process, protocol split. Running each method in its own process reproduces the multi-method BWT to within 9.6×10^{-5} , an order of magnitude below the pre-specified 10^{-3} threshold. This is a check on the protocol implementation, not on run-to-run variation of the model: two code paths that should agree do agree.

(b) Same session, arm order swapped. Within one D1 session the two arms run back to back, and reversing their order is a genuine perturbation of the run. At seed 42 the `gradient_B` arm’s BWT moves by 0.064 between the two orders (Appendix M); at seeds 789 and 2048 both arms stay

within 1.4×10^{-2} . This is a within-session positional effect, and the within-run contrast survives it at all three seeds.

(c) Cross-session, same seed. The absolute BWT of a fixed method and seed is not reproducible across sessions. For seed 42, `full_alora` records 0.491, 0.378, and 0.498 in three separate runs (spread 0.120), while `random_B-only` records 0.326, 0.324, and 0.278 (spread 0.048). The spread is method-dependent, so it does not act as a constant offset that cancels in a comparison.

Table 7: Absolute loss-based BWT for the same seed 42 under identical method definitions, across sessions and hardware. Run C is the D1 density-matched session and Run D the uniform-B suppression control (the M2 controller) session, both on a separate GPU and software stack; `random_A+B` was not run in either. A Run-D entry exists only for methods Run D ran, and it did not train a plain `full_alora` arm, so that cell is --- and its gradient-side arm is `gradient_B`, entered on its own row. Spread is $\max - \min$ across the runs in which the method appears.

Method	Run A	Run B	Run C	Run D / spread
Gradient mask (<code>full_alora</code>)	0.491	0.378	0.498	— (0.120)
Gradient-B@10% (<code>gradient_B</code>)	—	—	0.453	0.423
Random mask (<code>random_A+B</code>)	0.435	0.349	—	—
Random B-only (<code>random_B-only</code>)	0.326	0.324	0.278	0.278 (0.048)

Why this is not called NF4 non-determinism. Quantization is the same in every run of Table 7, so it cannot by itself explain a cross-session spread while leaving a same-process protocol split at 9.6×10^{-5} . The level-(c) spread is a property of the absolute BWT level of a session, not a random perturbation of the comparison. We therefore report it as *cross-session variation in the absolute BWT level* and do not attribute it to a mechanism. Two consequences follow. First, only within-run comparisons are used for any claim in this paper. Second, the `random_B-only` value that enters Table 2 is the D1 session’s own 0.278, taken from the same session as its partner arm, not an average across sessions.

Attribution of the level (c) spread. The level (c) spread has no single cause. Method position is not such a cause within a session: `set_seed` is called before each method’s model load and no random numbers are consumed between iterations, so the seed state entering training does not depend on position. Level (b) shows that position can still matter, and level (c) shows that session identity can. Which of these dominates across a full re-run is not tested here.

G RRM BOUNDARY ANALYSIS

We investigated whether assigning each task a different random mask (Rotating Random Masks, RRM) provides additional benefit beyond shared random masking. RRM generates a deterministic per-task mask via $m_i^{(t)} = \mathbf{1}[\text{hash}(t||i) \bmod 10 < k]$, reducing expected pairwise inter-task parameter overlap from 100% (shared mask) to k^2 (1% at 10% sparsity).

Table 8 reports RRM versus shared random across LoRA ranks and task order.

Table 8: RRM boundary analysis (Qwen2.5-3B, single seed per condition except rank-16 reference).

Condition	RRM BWT	Random BWT	Δ	Direction
Rank 4	0.464	0.423	+0.041	Random better
Rank 8	0.427	0.373	+0.055	Random better
Rank 16 (ref.)	0.401	0.435	−0.034	RRM better
Rank 32	0.414	0.390	+0.024	Random better
Reverse order	0.115	0.109	+0.006	Random better

RRM shows a favorable paired difference at rank 16 only. The overlap-reduction hypothesis predicts a consistent RRM advantage; the data do not support this across conditions. RRM’s effect appears condition-dependent rather than robust.

H SPARSITY SWEEP

We sweep the masking budget over 5%, 10%, 20%, and 30% (single seed) to test whether the within-pool ordering is specific to the 10% sparsity used in the main experiments. Across all four levels, shared random achieves numerically lower BWT than gradient attribution, with no cross-over; the largest margin is at 30%.

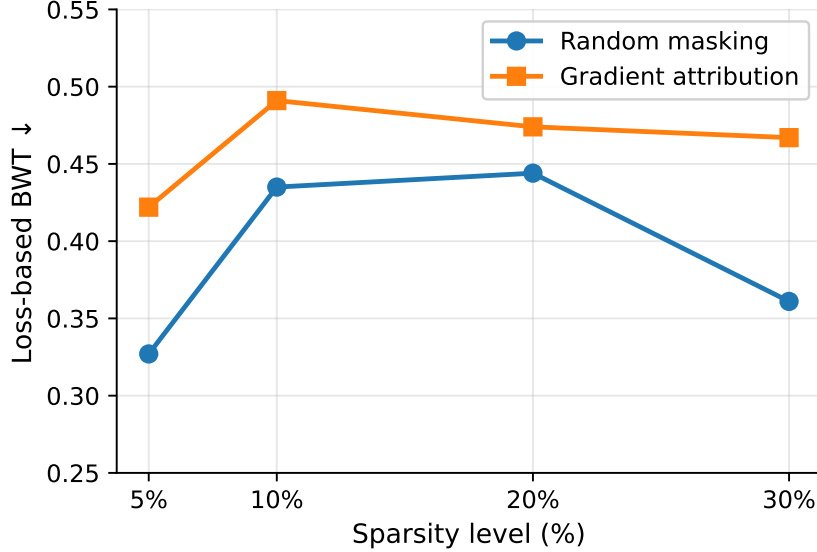


Figure 6: Sparsity sweep (Qwen2.5-3B, single seed). Shared random achieves numerically lower BWT than gradient attribution at all four tested levels.

Table 9: Sparsity sweep numerical values (Qwen2.5-3B, single seed). Random BWT $\leq$ Gradient BWT at all tested levels.

Sparsity	Random BWT ↓	Gradient BWT ↓
5%	0.327	0.422
10%	0.435	0.491
20%	0.444	0.474
30%	0.361	0.467

I TRAINING STABILITY: RANK-8 NO-MASK DIVERGENCE

The rank-8 no-mask ablation reveals a discrepancy between two endpoint-based forgetting summaries. Endpoint BWT is 0.176; average forgetting is 0.2837. Training produced a severe loss spike during task 4 (Math): evaluation losses on three prior tasks exceeded 14, compared to values below 4 for all masked conditions at the same rank. This spike subsequently recovered during task 5, yielding deceptively low endpoint BWT. We do not interpret the rank-8 no-mask endpoint BWT of 0.176 as evidence that no-mask outperforms masking.

The same signature recurs at rank 16 and is seed-dependent: at seed 123 the unmasked arm spiked during task 4 with evaluation losses of 16.2/16.0/14.7 on the three prior tasks (code/medical/legal), while every masked arm in that run stayed below 2.5; at seed 42 the unmasked arm did *not* spike (task-4 evaluations ≤ 1.2). The instability is therefore a property of the unmasked (all-parameter) configuration rather than of a particular rank, and it appears in a minority of seeds.

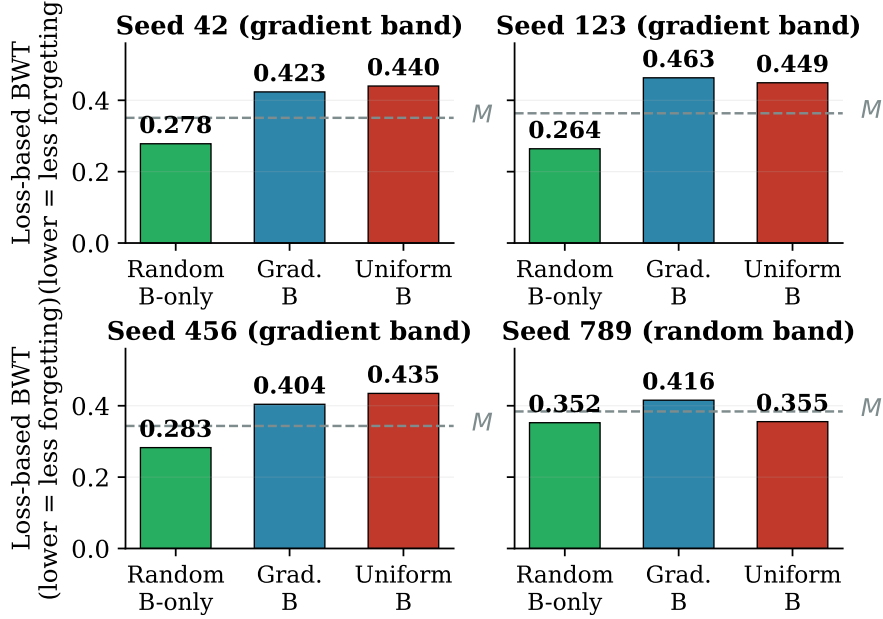


Figure 7: The suppression control’s outcome is *seed-dependent*. All three arms train 1,651,508 parameters, which is 5.5% of the full LoRA pool (each arm draws 10% of the `lora_B` sub-pool, and `B` holds 55% of the pool’s parameters), so this is the density-matched setting throughout. The uniform-B controller redistributes the gradient mask’s budget uniformly across LoRA modules. At seed 42 it sits in the gradient band (0.440 above gradient-B’s 0.423 and well above random-B-only’s 0.278); seeds 123 and 456 also land on the gradient side; at seed 789 it falls to the random band (0.355 below gradient-B’s 0.416, alongside random-B-only’s 0.352). Three of four seeds fall on the gradient side. M2 is an independent session from D1, so the same-seed gradient-B values differ between them (e.g. seed 42: 0.423 here vs 0.453 in the D1 table; seed 789: 0.416 vs 0.387; §F). We therefore do not treat this ablation as establishing the `o_proj`-concentration mechanism (§6).

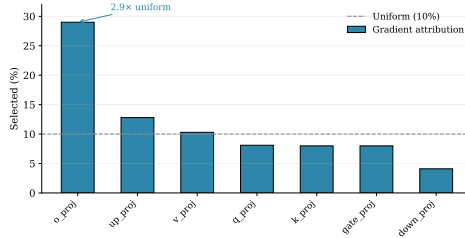


Figure 8: Gradient-mask concentration by LoRA module (Qwen2.5-3B). Gradient attribution selects 29% of `o_proj` parameters ($2.9\times$ the 10% uniform rate) and is under-represented in `down_proj` (4.1%); the dashed line marks uniform 10% selection.

J MASK CONCENTRATION BY MODULE

The `o_proj` suppression ablation and module-concentration figures, discussed in §6, are reproduced below together with the module-concentration table for reference.

K SCALE OBSERVATIONS

We additionally observe the within-pool comparison at three parameter scales (0.5B, 1.5B, 7B; single seed each). These are observational rather than multi-seed, and at 0.5B and 1.5B only the

Table 10: Gradient mask concentration by LoRA module (Qwen2.5-3B). Random masking distributes uniformly at 10% per module. Gradient attribution selects exclusively `lora_B` parameters (all `lora_A` receive zero importance due to $B = 0$ initialization).

Module	Gradient density	Random density	Ratio
o_proj	29.0%	10.0%	2.9×
up_proj	12.8%	10.0%	1.3×
v_proj	10.3%	10.0%	1.0×
q_proj	8.1%	10.0%	0.8×
k_proj	8.0%	10.0%	0.8×
gate_proj	8.0%	10.0%	0.8×
down_proj	4.1%	10.0%	0.4×

shared-random and no-mask conditions were run, so no gradient attribution baseline is available for a direct comparison there.

Table 11: BWT at additional scales (single seed, observational).

Scale	Random BWT ↓	Gradient BWT ↓	No Mask BWT ↓
0.5B	0.343	—	0.431
1.5B	0.309	—	0.356
7B	0.447	0.441	0.514

L TASK ORDERING (SINGLE SEED)

A reversed-order experiment yields $\text{BWT} \approx 0.11$ for all masking methods, compared to ≈ 0.42 under the standard order, a nearly $4\times$ reduction that far exceeds the method differences measured within this experiment (within-session method contrasts of order 10^{-2}). This is a *single seed*, so the $4\times$ magnitude is a qualitative pointer, not a quantitative estimate; it is reported to show that task ordering dominates the forgetting landscape, not as a measured effect size. The conclusions of this paper apply only to the tested Code→Medical→Legal→Math→Creative sequence; additional task orderings with multi-seed replication are needed before the null result can be generalized.

M METHOD ORDERING WITHIN A D1 SESSION (THREE SEEDS)

The D1 density-matched control runs two arms sequentially in one process: `random_B_only` then `gradient_B` (Run A), or the reverse (Run B). If method position within a session affects BWT, the within-run contrast (`random_B_only` – `gradient_B`) would change sign or magnitude when the order is reversed. We test this at three seeds (42, 789, 2048).

Design. Each seed produces two independent processes sharing the same model weights, LoRA configuration, and mask; only the execution order of the two arms differs. We report $\Delta_{\text{pos}}(\text{arm}) = \text{BWT}(\text{arm, position 1}) - \text{BWT}(\text{arm, position 2})$ for each arm and each seed.

Table 12: Method-ordering sensitivity (D1 density-matched, three seeds). Δ_{pos} is the BWT difference between running at position 1 and position 2. The last two columns express each arm’s positional shift as a fraction of the main D1 contrast ($|\Delta| = 0.155$).

Seed	$ \Delta_{\text{pos}}(\text{rB}) $	$ \Delta_{\text{pos}}(\text{gB}) $	rB / main	gB / main
42	1.5×10^{-3}	6.4×10^{-2}	0.01	0.41
789	3.1×10^{-5}	6.2×10^{-3}	0.00	0.04
2048	8.4×10^{-6}	1.3×10^{-2}	0.00	0.08

Results. Table 12 reports all three seeds. For the `random_B_only` arm the positional shift is negligible everywhere, at most 1.5×10^{-3} , which is one hundredth of the main contrast. For the `gradient_B` arm the shift is larger and seed-dependent: 0.064 at seed 42, which is 41% of the main contrast, but only 0.006 and 0.013 at seeds 789 and 2048, or 4% and 8% of it. Position therefore matters for one arm at one seed, and is otherwise small against the effect being measured.

Interpretation. These are three individual observations, not a distribution. At seed 42 the two readings remain open and inseparable at $n=1$: the Δ_{pos} values are either a real position effect or an artifact of treating a position offset as if it were the paired-comparison error, and the two later seeds favour the second reading. At seed 789 both arms recorded higher BWT at position 2 (Δ_{pos} negative for `gradient_B`), so a session-level shift is not excluded for that seed. At seed 2048 the arms moved in opposite directions, ruling out a pure session shift there.

The within-run contrast (`random_B_only` – `gradient_B`) is negative under both orders at all three seeds (seed 42: Run A = -0.121 , Run B = -0.186 ; seed 2048: Run A = -0.257 , Run B = -0.270), so the D1 direction survives order reversal.

N IN-DOMAIN CALIBRATION ABLATION

The primary experiments use out-of-domain Alpaca calibration data (64 samples). To test whether this choice drives the null result, we repeat the gradient-mask experiment with *pooled in-domain* calibration: 60 samples drawn equally from the five CL tasks.¹ All other parameters are identical (Qwen2.5-3B, seed 42, rank 16, 10% sparsity).

Loss-based BWT is 0.495 vs 0.491 with Alpaca calibration ($\Delta = 0.004$, 0.8% relative), well within the 5-seed standard deviation of 0.061. Random and no-mask controls are numerically identical across both runs, confirming infrastructure determinism. The *cross-calibration* overlap (Alpaca vs pooled) is Jaccard = 0.585: changing the calibration domain alters 42% of selected parameters. Despite this substantial mask difference, BWT changes by only 0.004. This robustness is of the *outcome*, not the mask: the mask does change, but *among gradient-derived masks* the specific 10% selected has little influence on forgetting, whereas swapping the selection criterion entirely (gradient for random) does, as the density-matched control shows (§5.1).

O B=0 RESTRICTION ASYMMETRY

The within-`lora_B` comparison varies the selection criterion while holding the parameter pool fixed. That the pool restriction is the right control to hold follows from an asymmetry between the two arms at seed 42 (Table 13). Restricting the gradient mask to `lora_B` changes nothing, because under $B = 0$ every `lora_A` importance score is already zero and the restriction is a no-op. The same restriction applied to the random arm removes the `lora_A` dilution and improves BWT by 0.025. A restriction that does nothing to one arm and helps the other is evidence that the two arms were drawing from different pools before the restriction was applied, which is what makes the pool-matched comparison the informative one.

Table 13: Effect of restricting the mask to `lora_B` (seed 42). For gradient attribution, the restriction is a no-op by construction ($B=0$ collapses every `lora_A` importance score to zero); for random, restricting to B removes `lora_A` dilution and improves BWT by 0.025. This asymmetry demonstrates that the within-B comparison varies the selection criterion, not the parameter pool.

Method	Unrestricted BWT	B-only BWT	Gap
Gradient	0.378	0.378	+0.000
Random	0.349	0.324	−0.025

¹ $\lfloor 64/5 \rfloor \times 5 = 60$; a 6.25% size difference from integer division.

Table 14: Per-seed results, both columns in the direction [Grad – Random] (positive = gradient forgets more). Col. 2 (accuracy difference) is measured on an A800 GPU; Col. 3 (loss-based BWT difference) is measured on the same A800 GPU. Taken together they form a second environment relative to the original V100 loss runs. Col. 2 carries the 95% t -interval on the seven-seed mean. Col. 3: mean +0.152 (SD 0.0749, $t = 5.37$, CI [+0.083, +0.221], 7/7 same direction). The abstract’s -0.155 is the same loss contrast measured on the V100 stack under the opposite convention [Random – Grad].

Seed	Accuracy Δ [Grad – Rand]	Loss BWT Δ [Grad – Rand]
42	+0.0200	+0.2320
123	0.0000	+0.1647
456	+0.0048	+0.1300
789	−0.0698	+0.0786
1024	+0.0460	+0.0514
2048	+0.0051	+0.2572
4096	+0.0600	+0.1492
<i>Summary (column 2 in percentage points, column 3 in BWT units)</i>		
Mean	+0.94 pp	+0.152
SD	4.17 pp	0.0749
95% CI	[−2.91, +4.80] pp	[+0.083, +0.221]

Table 15: Cross-architecture comparison on Qwen2.5-3B and Phi-2 (2.78B, GPT-NeoX-style), both arms within a row sharing one target set. *Top*: confounded design, gradient selecting B-only and random selecting A+B; both architectures use all linear projections ($q, k, v, fc1, fc2, dense$; pool 23,592,960). *Bottom*: within-lora_B design, both methods restricted to lora_B; the Phi-2 rows use a *different* target set omitting dense ($q, k, v, fc1, fc2$; pool 20,971,520, 12.5% smaller), so the two designs are not comparable row-for-row. The within-lora_B rows are matched on pool but *not* density (the gradient mask activates approximately $2\times$ the B parameters of Random-B), as on Qwen2.5-3B.

Architecture	Comparison	Grad. BWT $\downarrow$	Rand. BWT $\downarrow$	Δ
Qwen2.5-3B ($n=5$)	Confounded	0.437 ± 0.061	0.428 ± 0.019	−0.009
Phi-2 ($n=5$)	Confounded	0.158 ± 0.019	0.183 ± 0.016	+0.025
Qwen2.5-3B ($n=7$)	Within-B	0.409 ± 0.046	0.338 ± 0.015	−0.072
Phi-2 ($n=4$)	Within-B	0.149 ± 0.036	0.065 ± 0.017	−0.084

P PER-SEED ACCURACY AND MATCHED LOSS CONTRAST

The accuracy evaluation is reported in the main text, §5.4, together with the option-scoring protocol; its per-seed values and the matched loss contrast are tabulated below.

Q CROSS-ARCHITECTURE COMPARISON

Both cross-architecture designs are tabulated together below: the confounded design (gradient selecting B-only, random selecting A+B) and the within-lora_B design (both methods restricted to lora_B). The two designs are not comparable row-for-row because the Phi-2 within-B rows use a different LoRA target set; §6.1 discusses both.

R THE MATH TASK: TRUNCATION AUDIT

Loss on math decreases with training while accuracy on the same task collapses to zero for all arms: peak accuracy at math’s own training phase is 0.04 and every arm ends at 0.00. The cause is not a format mismatch—the numeric-match function accepts a bare number, a number with surrounding text, and a complete solution alike, and the training and evaluation prompt prefixes are identical by construction. The surviving explanation is truncation: a generation audit reports 56–95% of gener-

ations across all tasks and arms reaching the 256-token cap without an end-of-sequence token, and for math specifically 81–90% of mathematical generations, accumulated across the six evaluation calls per task, reach the cap the same way. The last number in a truncated generation is an intermediate quantity, so the reference answer is absent from the value the scorer inspects. Every arm shares the same cap and scorer, so the bias is common rather than arm-specific. We do not treat the math column as a forgetting result: under option-based scoring math yields no score at all, and under generation scoring it is confounded with truncation. Scoring each option by its log-likelihood removes the truncation dependence, and that is the scorer used above.

S LIMITATIONS

- *Static masks only.* All methods reuse a single static mask across all five tasks. Whether adaptive per-task gradient masks (recomputed after each task’s training) would change the outcome remains an open question; this is the broadest scope limitation of the present study.
- *Absolute BWT is not reproducible across sessions.* The same method and seed produces absolute BWT values that vary by up to ~ 0.12 between sessions, a spread that approaches the within-B advantage ($\Delta = 0.072$). This is not a random perturbation of a comparison and we do not attribute it to a mechanism; it is a property of the absolute level a session settles at (Appendix F separates this from two other, smaller run-to-run quantities). The consequence is scope, not invalidity: *only within-run comparisons are valid*, and every method comparison in this paper is within-run, sharing one RNG state and one method ordering.
- *B-pool advantage is not robust.* Restricting the random mask to the `lora_B` pool (`random_B_only`) does not consistently outperform the unrestricted random mask (`random_A+B`); the direction is seed-dependent and the mean difference is small relative to the cross-session spread, so this is not a systematic effect. The pool restriction therefore acts as a supporting observation in this paper; the density-matched comparison carries the headline claim.²
- *Loss-based evaluation with an accuracy consistency check.* All multi-seed statistical tests are loss-based BWT. An accuracy evaluation over the same seven seeds, on the two discrete-option tasks (Medical and Legal, 100 items each), returns a mean gradient-minus-random difference of +0.94 percentage points with a 95% interval of $[-2.91, +4.80]$ percentage points. The two masks are indistinguishable on accuracy. A 4-percentage-point non-inferiority margin was specified in advance for this interval and is not met (upper bound +4.80 percentage points). Accuracy differences of up to about 4.8 percentage points in the gradient mask’s favour are not excluded by this measurement. See §5.4. Conclusions are scoped to the loss metric for all multi-seed claims.
- *Calibration domain tested but not exhaustively.* A pooled in-domain calibration ablation (single seed) changed 42% of selected parameters yet produced near-identical BWT ($\Delta = 0.004$; §N). The remaining untested variant is per-task calibration (a separate mask per task).
- *Single task order.* The fixed Code→Medical→Legal→Math→Creative order is the primary condition. A reversed-order experiment (single seed) yields BWT ≈ 0.11 for all methods, a nearly 4× reduction; conclusions are specific to the tested sequence.
- *Statistical power ($n=7$).* With seven seeds the within-B comparison is significant on both parametric and non-parametric tests (paired $t(6) = -3.53$, $p = 0.012$; Wilcoxon $p = 0.016$), with 7/7 seeds directionally consistent and a large effect size ($d_z = -1.34$). The 7-seed paired test uses within-run differences, which are unaffected by the cross-session spread; the reported means (0.338 vs 0.409) are cross-session averages and carry ± 0.12 uncertainty.
- *Single LoRA rank.* Multi-seed results are at rank 16 only; other ranks are single-seed.
- *Zero-init LoRA only.* The B=0 mechanism and within-B comparison apply only to standard zero-init LoRA. PiSSA (Meng et al., 2024) and other non-zero initialization variants are not tested.

²Within-B vs gradient attribution is the headline (7/7 consistent, $p = 0.012$); within-B vs random-A+B is seed-dependent.

T POSITIONING AGAINST GRADIENT-ATTRIBUTION FAMILIES

Table 16 places our contribution against the gradient-attribution families we are positioned against, on the two axes that drive the study.

Table 16: Positioning against gradient-attribution families. "B=0 pool" isolates the `lora_B` pool; "within-pool grad-vs-random" compares gradient to a matched random baseline in a shared pool.

Method	B=0 pool	Within-pool grad-vs-random
EWC / SI / MAS	No	No
PiSSA ($B \neq 0$)	N/A by init	No
Attribution-Guided CL	No	No
Ours (within-<code>lora_B</code>)	Yes	Yes

Table 17 lists the selection rules this paper is positioned against and marks which we train under one protocol. Only the rules we train contribute empirical numbers here; the rest are compared by their published design, with none of their reported numbers presented as a measurement of ours.

Table 17: Selection rules this paper is positioned against. The last column marks the four rules we train and evaluate under one protocol (§3.3); the rest are compared by design only, and none of their published numbers is cited as a result of ours.

Rule	What it scores, and what for	Run
EWC (Kirkpatrick et al., 2017); MAS (Aljundi et al., 2018)	Fisher information, output sensitivity; a penalty, not a mask	no
SNIP (Lee et al., 2019); GraSP (Wang et al., 2020)	gradient at initialisation; one-shot pruning mask	no
Wanda (Sun et al., 2024)	weight $\times$ activation magnitude; one-shot pruning mask	no
LoRA-FA (Zhang et al., 2023a)	none, structural; freezes <code>lora_A</code> , trains <code>lora_B</code>	no
Grad-Full, Grad-B (§3.3)	calibration gradient magnitude; static CL mask, full pool (Grad-Full) or <code>lora_B</code> at 10% (Grad-B)	yes
Random-B; Random A+B (Xu & Zhang, 2024)	none, uniform; static CL mask, <code>lora_B</code> at 10% (Random-B) or full pool (A+B)	yes

U D1 PER-TASK AND PER-SEED DECOMPOSITION

Table 18 gives the seven-seed mean per-task contribution. Table 19 gives the same quantity for each seed, so the task-level pattern can be checked against the seed-level one; the main text reports the same decomposition as a figure (Figure 3). Each entry is $\text{loss_matrix}[4][j] - \text{loss_matrix}[j][j]$ for prior task j ; the final column is the mean over the four tasks, which is the per-seed aggregate Δ of Table 2.

V RULING OUT A LEARNING DEFICIT IN THE D1 GAP

The D1 density-matched comparison (both arms 10% of `lora_B`, 1,651,508 parameters) shows random-B-only achieving lower BWT than gradient-B. One rival reading is that gradient-B's top-10% is the more concentrated half of the original 20% mask, so gradient-B learns each task less well and the gap is a learning deficit rather than a retention difference. The per-task diagonal losses (each task's loss immediately after it is learned) exclude this: gradient-B's diagonal loss is lower than random-B-only's in *all* 35 seed $\times$ task cells (7 seeds $\times$ 5 tasks), with no exceptions.

Table 18: D1 per-task decomposition of loss-based BWT, averaged over the seven seeds. Each entry is $\text{loss_matrix}[4][j] - \text{loss_matrix}[j][j]$ for prior task j , that task’s contribution to the aggregate; the last column is its share of the aggregate Δ in absolute value. Creative is the final task and contributes no forgetting term. The aggregate is concentrated on medical (79.9%) but not confined there: excluding medical the matched contrast remains -0.042 (7/7 negative), while code and math are each negative in 7/7 seeds and legal is not established.

Prior task	random-B	gradient-B	Δ	share of Δ
Code	-0.021	0.025	-0.046	7.4%
Medical	0.998	1.493	-0.496	79.9%
Legal	0.007	0.010	-0.003	0.5%
Math	0.191	0.266	-0.076	12.2%
Aggregate	0.294	0.449	-0.155	100%

Table 19: Per-seed, per-task D1 decomposition ($\Delta = \text{random-B} - \text{gradient-B}$ for each prior task). Negative favours random-B. Legal is the only task whose sign is not consistent across seeds.

Seed	Code	Medical	Legal	Math	Aggregate
42	-0.040	-0.409	-0.002	-0.249	-0.175
123	-0.002	-0.698	-0.002	-0.052	-0.188
456	-0.074	-0.381	-0.004	-0.033	-0.123
789	-0.063	-0.054	+0.001	-0.023	-0.035
1024	-0.044	-0.410	-0.011	-0.021	-0.121
2048	-0.041	-0.950	+0.010	-0.062	-0.261
4096	-0.059	-0.568	-0.014	-0.091	-0.183
Mean	-0.046	-0.496	-0.003	-0.076	-0.155

Table 20: Diagonal loss (mean $\pm$ std over the seven D1 seeds), by task. Lower is better. gradient-B is lower than random-B-only in every cell (Table 20); Grad-Full is the uncontrolled reference. Loss diagonals only; the D1 run records contain no accuracy matrix (accuracy is measured separately in the accuracy branch, §5.4).

Task	random_B_only	gradient_B	full_alora
code	0.952 ± 0.111	0.901 ± 0.100	0.898 ± 0.099
medical	1.190 ± 0.040	1.152 ± 0.040	1.144 ± 0.035
legal	0.388 ± 0.035	0.368 ± 0.038	0.365 ± 0.037
math	3.838 ± 0.103	3.652 ± 0.096	3.642 ± 0.100
creative	2.104 ± 0.053	2.067 ± 0.053	2.056 ± 0.054

A second rival reading is that BWT, being a difference of losses, is inflated for gradient-B simply because it starts from a lower per-task loss. Reconstructing each seed’s mean per-task endpoint (BWT plus the mean diagonal over the first four tasks) closes this from both sides: gradient-B starts *lower* yet ends *higher* in 6 of 7 seeds, so its larger BWT reflects greater forgetting rather than a lower baseline (Table 21).

Both checks are loss-based (accuracy is measured separately in the accuracy branch, §5.4), use a single task order, and cover Qwen2.5-3B only. The endpoint reconstruction is a derived quantity whose per-seed differences are of the same magnitude as the cross-run drift documented in Appendix F; it is therefore offered as a qualitative directional exclusion, not a significance claim.

W CONFOUNDED COMPARISON TABLE

Table 22 gives the full confounded-design comparison discussed in §5.1. The design varies pool and criterion together, so it is reported for completeness rather than as evidence about the selection rule.

Table 21: Derived per-task endpoint loss (BWT + mean diagonal over the first four tasks), per seed. gradient-B’s endpoint is higher in 6 of 7 seeds; seed 789 is the exception. These are derived quantities of the same order as cross-run drift, reported for the qualitative direction only.

Seed	rB start	rB end	gB start	gB end (diff)
42	1.543	1.821	1.478	1.931 (+0.110)
123	1.632	1.896	1.553	2.006 (+0.110)
456	1.575	1.858	1.494	1.900 (+0.042)
789	1.548	1.900	1.475	1.862 (−0.038)
1024	1.566	1.918	1.489	1.962 (+0.045)
2048	1.665	1.896	1.587	2.078 (+0.182)
4096	1.618	1.913	1.553	2.031 (+0.118)

Table 22: Confounded comparison (Qwen2.5-3B, rank-16 LoRA, 5 tasks, $n=5$ seeds). Incremental cost is the attribution pass’s transient peak (~ 0.73 GB; 0 for random by construction; Appendix B). Across all tables, bold marks the better value in its column.

Method	BWT ↓	Loss ↓	Incremental cost
Shared random	0.428 ± 0.019	1.940 ± 0.021	0
Gradient mask	0.437 ± 0.061	1.966 ± 0.037	~ 0.73 GB
No mask	0.484 ± 0.047	1.970 ± 0.033	0